

DESIGN AND ANALYSIS OF ALGORITHM

WEEK 5

Q)1

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main(){
4     int n;
5     scanf("%d", &n);
6
7     int a[n];
8
9     for(int i=0;i<n;i++){
10         scanf("%d", &a[i]);
11     }
12
13     for(int i=0;i<n;i++){
14         for(int j=i+1;j<n;j++){
15             if(a[i] == a[j]){
16                 printf("%d", a[i]);
17                 return 0;
18             }
19         }
20     }
21     return 0;
22 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Q)2

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0%)

```
1 #include <stdio.h>
2
3 int main(){
4     int n;
5     scanf("%d", &n);
6
7     int a[n];
8
9     for(int i=0;i<n;i++){
10         scanf("%d", &a[i]);
11     }
12     int slow = a[0];
13     int fast = a[0];
14
15     do{
16         slow = a[slow];
17         fast = a[a[fast]];
18     }while (slow != fast);
19     slow = a[0];
20     while(slow != fast){
21         slow = a[slow];
22         fast = a[fast];
23     }
24     printf("%d", slow);
25     return 0;
26 }
27
28 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Q)3

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1	10 57
3 10 17 57	
6	
2 7 10 15 57 246	

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main(){
4     int T;
5     scanf("%d", &T);
6
7     while(T--){
8         int n1, n2;
9         scanf("%d", &n1);
10
11         int a[n1];
12         for(int i=0; i<n1; i++){
13             scanf("%d", &a[i]);
14         }
15         scanf("%d", &n2);
16
17         int b[n2];
18         for(int i=0; i<n2; i++){
19             scanf("%d", &b[i]);
20         }
21         int first = 1;
22
23         for(int i=0; i<n1; i++){
24             for(int j=0; j<n2; j++){
25                 if(a[i] == b[j]){
26                     if(!first)
27                         printf(" ");
28                     printf("%d", a[i]);
29                     first = 0;
30                     break;
31                 }
32             }
33         }
34         printf("\n");
35
36     }
```

Q)4

Output Format
The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1	10 57
2 10 17 57	
6	
2 7 10 15 57 246	

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main(){
4     int T;
5     scanf("%d", &T);
6
7     while(T--){
8         int n1, n2;
9         scanf("%d", &n1);
10
11         int a[n1];
12         for(int i=0; i<n1; i++){
13             scanf("%d", &a[i]);
14         }
15         scanf("%d", &n2);
16
17         int b[n2];
18         for(int i=0; i<n2; i++){
19             scanf("%d", &b[i]);
20         }
21         int i=0, j=0, first=1;
22         while(i<n1 && j<n2){
23             if(a[i]<b[j]){
24                 i++;
25             }
26             else if (a[i] > b[j]){
27                 j++;
28             }
29             else if (first){
30                 printf(" ");
31                 printf("%d", a[i]);
32                 first = 0;
33                 i++;
34                 j++;
35             }
36         }
37         printf("\n");
38     }
39     return 0;
40 }
41 }
```

	Input	Expected	Got	
✓	1	10 57	10 57	✓

Q)5

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[i] - A[j] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3	1
1 3 5	
4	

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main(){
4     int n,k;
5     scanf("%d",&n);
6
7     int a[n];
8
9     for(int i=0;i<n;i++){
10         scanf("%d",&a[i]);
11
12         scanf("%d",&k);
13
14         for(int i=0;i<n;i++){
15             for(int j=i+1;j<n;j++){
16                 if (a[j] - a[i] == k){
17                     printf("1");
18                     return 0;
19                 }
20             }
21         }
22         printf("0");
23
24         return 0;
25 }
```

Input	Expected	Got
3 1 3 5 4	1	1 ✓
10 1 4 6 8 12 14 15 20 21 25 1	1	1 ✓
10 1 2 3 5 11 14 16 24 28 29 0	0	0 ✓
10 0 2 3 7 13 14 15 20 24 25 10	1	1 ✓

Passed all tests! ✓

Correct

Q)6

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[i] - A[j] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
5	1
1 3 5	
4	

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 int main(){
4     int n,k;
5     scanf("%d",&n);
6
7     int a[n];
8
9     for(int i=0;i<n;i++){
10         scanf("%d",&a[i]);
11     }
12     scanf("%d",&k);
13
14     int i=0, j=1;
15
16     while(i<n && j<n){
17         if(i==j){
18             j++;
19         }else if (a[j] - a[i] == k){
20             printf("1");
21             return 0;
22         }else if(a[j] - a[i] < k){
23             j++;
24         }else{
25             i++;
26         }
27     }
28     printf("0");
29     return 0;
30 }

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 16 20 24 25	1	1	✓